

MAT 102: Ordinary Differential Equations

Topic 2: First Order Differential Equations

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1 Linear First Order Equations

1.1 Definition and Standard Form

A first-order ODE is **linear** if it can be written as:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where $P(x)$ and $Q(x)$ are functions of x only (not of y).

- If $Q(x) = 0$: the equation is **homogeneous linear** $\rightarrow$ separable.
- If $Q(x) \neq 0$: the equation is **non-homogeneous linear**.

1.2 Method of Integrating Factor

The key idea: multiply through by a special function $\mu(x)$ so that the left-hand side becomes an exact derivative $\frac{d}{dx}[\mu y]$.

Algorithm — Integrating Factor Method

Step 1. Write the equation in standard form: $\frac{dy}{dx} + P(x)y = Q(x)$

Step 2. Compute $\mu(x) = e^{\int P(x) dx}$ (no constant of integration needed here)

Step 3. Multiply both sides by $\mu(x)$:

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

Step 4. Integrate both sides:

$$\mu(x)y = \int \mu(x)Q(x) dx + C$$

Step 5. Solve for y :

$$y = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x) dx + C \right]$$

Why does this work? Notice that $\mu'(x) = \mu(x)P(x)$, so:

$$\mu y' + \mu P(x)y = \frac{d}{dx}[\mu y]$$

This is exactly the product rule applied to μy .

Example 1: Solve $\frac{dy}{dx} - 2y = 4x$.

Standard form: $P(x) = -2$, $Q(x) = 4x$.

$$\mu = e^{\int -2 dx} = e^{-2x}$$

$$\frac{d}{dx}[e^{-2x}y] = 4xe^{-2x}$$

$$e^{-2x}y = \int 4xe^{-2x} dx$$

Using integration by parts ($u = 4x$, $dv = e^{-2x}dx$):

$$= -2xe^{-2x} + \int 2e^{-2x} dx = -2xe^{-2x} - e^{-2x} + C$$

$$\boxed{y = -2x - 1 + Ce^{2x}}$$

Example 2: Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$, $x > 0$.

$$P(x) = \frac{1}{x}, Q(x) = x^2.$$

$$\mu = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

$$\frac{d}{dx}[xy] = x \cdot x^2 = x^3$$

$$xy = \int x^3 dx = \frac{x^4}{4} + C$$

$$\boxed{y = \frac{x^3}{4} + \frac{C}{x}}$$

Example 3: Solve $\frac{dy}{dx} + 2xy = 2x$.

$$P(x) = 2x, Q(x) = 2x.$$

$$\mu = e^{\int 2x dx} = e^{x^2}$$

$$\frac{d}{dx}[e^{x^2}y] = 2xe^{x^2}$$

$$e^{x^2}y = \int 2xe^{x^2} dx = e^{x^2} + C$$

$$\boxed{y = 1 + Ce^{-x^2}}$$

Note: as $x \rightarrow \infty$, $y \rightarrow 1$ for any C — the equilibrium solution is $y = 1$.

Example 4: Solve $x \frac{dy}{dx} - 3y = x^4$, $x > 0$.

Divide by x : $\frac{dy}{dx} - \frac{3}{x}y = x^3$.

$P(x) = -\frac{3}{x}$, $Q(x) = x^3$.

$$\mu = e^{\int -3/x dx} = e^{-3 \ln x} = x^{-3}$$

$$\frac{d}{dx}[x^{-3}y] = x^{-3} \cdot x^3 = 1$$

$$x^{-3}y = x + C \quad \Rightarrow \quad \boxed{y = x^4 + Cx^3}$$

2 Separable Equations

2.1 Definition

A first-order ODE is **separable** if it can be written so that all terms involving y (and dy) are on one side, and all terms involving x (and dx) are on the other:

$$\frac{dy}{dx} = f(x) g(y) \quad \Leftrightarrow \quad \frac{dy}{g(y)} = f(x) dx \quad (g(y) \neq 0)$$

Algorithm — Separation of Variables

Step 1. Rewrite as $\frac{dy}{g(y)} = f(x) dx$

Step 2. Integrate both sides: $\int \frac{dy}{g(y)} = \int f(x) dx + C$

Step 3. Solve for y explicitly if possible, otherwise leave implicitly.

Step 4. Check for any solutions lost when dividing by $g(y)$ (i.e., where $g(y) = 0$).

Example 1: Solve $\frac{dy}{dx} = xy$.

$$\frac{dy}{y} = x dx$$

$$\int \frac{dy}{y} = \int x \, dx \quad \Rightarrow \quad \ln |y| = \frac{x^2}{2} + C_1$$

$$|y| = e^{C_1} e^{x^2/2} \quad \Rightarrow \quad \boxed{y = A e^{x^2/2}}, \quad A \in \mathbb{R}$$

(Note: $A = 0$ gives the trivial solution $y = 0$, which was lost when dividing by y .)

Example 2: Solve $\frac{dy}{dx} = \frac{x^2}{1-y^2}$.

$$(1-y^2) \, dy = x^2 \, dx$$

$$\int (1-y^2) \, dy = \int x^2 \, dx$$

$$y - \frac{y^3}{3} = \frac{x^3}{3} + C$$

$$\boxed{3y - y^3 = x^3 + C} \quad (\text{implicit solution})$$

Example 3: Solve $(1+x) \, dy - y \, dx = 0$.

$$\frac{dy}{y} = \frac{dx}{1+x}$$

$$\ln |y| = \ln |1+x| + C_1$$

$$\boxed{y = A(1+x)}, \quad A \neq 0$$

Example 4: Solve $\frac{dy}{dx} = e^{x-y}$ with $y(0) = 0$.

$$\frac{dy}{dx} = e^x \cdot e^{-y}$$

$$e^y \, dy = e^x \, dx$$

$$e^y = e^x + C$$

Apply $y(0) = 0$: $e^0 = e^0 + C \Rightarrow C = 0$.

$$e^y = e^x \quad \Rightarrow \quad \boxed{y = x}$$

Example 5: Radioactive decay. A substance decays at rate proportional to amount present. If initially N_0 grams, find $N(t)$.

$$\frac{dN}{dt} = -\lambda N \quad \Rightarrow \quad \frac{dN}{N} = -\lambda dt$$

$$\ln N = -\lambda t + C \quad \Rightarrow \quad \boxed{N(t) = N_0 e^{-\lambda t}}$$

3 Homogeneous Differential Equations

3.1 Definition

A first-order ODE $\frac{dy}{dx} = f(x, y)$ is called **homogeneous** if $f(x, y)$ can be expressed purely as a function of the ratio $\frac{y}{x}$:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

This means the right side depends only on the ratio y/x , not on x and y separately

Test: An ODE $M dx + N dy = 0$ is homogeneous if M and N are both **homogeneous functions of the same degree**, i.e., $M(tx, ty) = t^n M(x, y)$ and $N(tx, ty) = t^n N(x, y)$.

$M dx + N dy = 0$ is considered **homogeneous** if you can scale both variables by a factor t and “pull out” the same power of t from both functions

Algorithm — Homogeneous Substitution

Step 1. Confirm the equation is homogeneous, that is check if the equation can be written as:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

Step 2. Use substitution. Let $v = \frac{y}{x}$, so $y = vx$

Then differentiate:

$$\frac{dy}{dx} = v + x \frac{dv}{dx}.$$

Step 3. Substitute into original equation to get a separable equation in v and x .

Step 4. Separate and integrate.

Step 5. Back-substitute $v = y/x$.

Example 1: Solve $\frac{dy}{dx} = \frac{x+y}{x}$

step1: $\frac{dy}{dx} = 1 + \frac{y}{x}$ (1) (homogeneous)

Step2: Let $v = \frac{y}{x}$ implies $y = vx$, then $\frac{dy}{dx} = v + x \frac{dv}{dx}$

step3: Replace $y = vx$, $y/x = v$, and $dy/dx = v + xdv/dx$ into(1) to get

$$v + x \frac{dv}{dx} = 1 + v$$

Simplify

$$\frac{dv}{dx} = \frac{1}{x}$$

Separate variables and integrate both sides:

$$\int dv = \int \frac{1}{x} dx$$

$$v = \ln |x| + C$$

Replace $v = y/x$ and simplify:

$$y = x(\ln |x| + C)$$

Example 2: Solve $\frac{dy}{dx} = \frac{y}{x} + \frac{x}{y}$.

Let $v = y/x$, $y = vx$:

$$v + x \frac{dv}{dx} = v + \frac{1}{v}$$

$$x \frac{dv}{dx} = \frac{1}{v}$$

$$v \, dv = \frac{dx}{x}$$

$$\frac{v^2}{2} = \ln |x| + C$$

$$\frac{y^2}{2x^2} = \ln |x| + C \quad \Rightarrow \quad \boxed{y^2 = 2x^2(\ln |x| + C)}$$

Example 3: Solve $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$.

Let $v = y/x$:

$$v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{v^2 - 1 - 2v^2}{2v} = \frac{-v^2 - 1}{2v}$$

$$\frac{2v \, dv}{v^2 + 1} = -\frac{dx}{x}$$

$$\ln(v^2 + 1) = -\ln |x| + C$$

$$(v^2 + 1)x = A \quad \Rightarrow \quad \boxed{y^2 + x^2 = Ax}$$

Example 4: Solve $(x^2 + y^2) \, dx - 2xy \, dy = 0$.

Check: $M = x^2 + y^2$ (degree 2), $N = -2xy$ (degree 2). Homogeneous.

Let $v = y/x$:

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy} = \frac{1 + v^2}{2v}$$

$$v + x \frac{dv}{dx} = \frac{1 + v^2}{2v}$$

$$x \frac{dv}{dx} = \frac{1 + v^2 - 2v^2}{2v} = \frac{1 - v^2}{2v}$$

$$\frac{2v dv}{1 - v^2} = \frac{dx}{x}$$

$$-\ln|1 - v^2| = \ln|x| + C_1$$

$$(1 - v^2)x = A \Rightarrow x \left(1 - \frac{y^2}{x^2} \right) = A \Rightarrow \boxed{x^2 - y^2 = Ax}$$

4 Bernoulli Equation

4.1 Definition

The **Bernoulli equation** is:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \quad n \in \mathbb{R} \neq 0, 1$$

- When $n = 0$: reduces to a linear equation.
- When $n = 1$: reduces to a separable equation.
- For $n \neq 0, 1$: requires the Bernoulli substitution.

Algorithm — Bernoulli Substitution

Step 1. Divide both sides by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

Step 2. Let $v = y^{1-n}$. Then $\frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx}$, so $y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dv}{dx}$.

Step 3. The equation becomes **linear** in v :

$$\frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)$$

Step 4. Solve using the integrating factor method.

Step 5. Back-substitute $v = y^{1-n}$.

Example 1: Solve $\frac{dy}{dx} + y = xy^3$.

$n = 3, P = 1, Q = x$. Divide by y^3 :

$$y^{-3} \frac{dy}{dx} + y^{-2} = x \dots (*)$$

Use Substitution, let $v = y^{1-3} = y^{-2}$, $y^{-3} dy/dx = -1/2 dv/dx$. Substitute to $\dots (*)$

$$\frac{dv}{dx} - 2v = -2x$$

Integrating factor: $\mu = e^{-2x}$

$$\frac{d}{dx}[e^{-2x}v] = -2xe^{-2x}$$

$$e^{-2x}v = \int -2xe^{-2x} dx$$

$$\text{Let } u = -2x \Rightarrow du = -2dx. \quad \text{Let } dv = e^{-2x}dx \Rightarrow v = \int e^{-2x}dx = \frac{1}{2}e^{-2x}$$

$$e^{-2x}v = \int -2xe^{-2x} dx = xe^{-2x} + \frac{1}{2}e^{-2x} + C$$

$$v = x + \frac{1}{2} + Ce^{2x} \quad \Rightarrow \quad \boxed{y^{-2} = x + \frac{1}{2} + Ce^{2x}}$$

Example 2: Solve $\frac{dy}{dx} - \frac{y}{x} = xy^2$.

$n = 2, P = -1/x, Q = x$. Let $v = y^{-1}$:

$$\frac{dv}{dx} + \frac{v}{x} = -x$$

$$\mu = e^{\int 1/x dx} = x$$

$$\frac{d}{dx}[xv] = -x^2$$

$$xv = -\frac{x^3}{3} + C \quad \Rightarrow \quad v = -\frac{x^2}{3} + \frac{C}{x}$$

$$\boxed{\frac{1}{y} = \frac{C}{x} - \frac{x^2}{3}}$$

Example 3: Solve $3\frac{dy}{dx} + \frac{y}{x} = \frac{12}{y^2}$, $x > 0$.

Rewrite: $\frac{dy}{dx} + \frac{y}{3x} = \frac{4}{y^2}$.

$n = -2$, so $v = y^{1-(-2)} = y^3$, $\frac{dv}{dx} = 3y^2 \frac{dy}{dx}$.

Multiply original by $3y^2$:

$$3y^2 \frac{dy}{dx} + \frac{y^3}{x} = 12 \quad \Rightarrow \quad \frac{dv}{dx} + \frac{v}{x} = 12$$

$\mu = x$:

$$\frac{d}{dx}[xv] = 12x \quad \Rightarrow \quad xv = 6x^2 + C$$

$$\boxed{y^3 = 6x + \frac{C}{x}}$$

5 Exact Differential Equations and Integrating Factors

Definition — Exact Equation

The equation $M(x, y) dx + N(x, y) dy = 0$ is **exact** on a region R if there exists a function $F(x, y)$ such that:

$$\frac{\partial F}{\partial x} = M(x, y) \quad \text{and} \quad \frac{\partial F}{\partial y} = N(x, y)$$

The general solution is then $F(x, y) = C$.

Test for Exactness: If M , N , M_y , N_x are continuous on R , then the equation is exact if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Why? If F exists, then $F_{xy} = M_y$ and $F_{yx} = N_x$. By equality of mixed partials (Clairaut's theorem), $M_y = N_x$.

Algorithm — Solving Exact Equations

Step 1. Check $M_y = N_x$. If not equal $\rightarrow$ not exact (try integrating factor).

Step 2. Integrate M with respect to x :

$$F(x, y) = \int M(x, y) dx + h(y)$$

Step 3. Differentiate F with respect to y and set equal to N :

$$\frac{\partial F}{\partial y} = N(x, y) \Rightarrow \text{solve for } h'(y)$$

Step 4. Integrate $h'(y)$ to find $h(y)$.

Step 5. Write the general solution: $F(x, y) = C$.

Example 1: Solve $(2xy + 3) dx + (x^2 + 4y) dy = 0$. **Step 1: Identify** $M = 2xy + 3$, $N = x^2 + 4y$.

Step 2: Check exactness (Differentiate) $M_y = 2x$, $N_x = 2x$. Exact.

Step 3: Integrate M with respect to x , treat y as constant and add unknown function of y

$$F = \int (2xy + 3) dx = x^2y + 3x + h(y)$$

Step 4: Differentiate F wrt y and set equal to N to get the unknown function of y .

$$F_y = x^2 + h'(y) = x^2 + 4y \Rightarrow h'(y) = 4y$$

Step 5: Integrate to get the unknown function

$$h = 2y^2$$

Final answer:

$$F = \boxed{x^2y + 3x + 2y^2 = C}$$

>If the differential of a function is zero, then the function is not changing. So its value stays constant. Hence: $F(x, y) = C$, where C is a constant.

Example 2: Solve $(e^x \sin y - 2y \sin x) dx + (e^x \cos y + 2 \cos x) dy = 0$.

$$M = e^x \sin y - 2y \sin x, N = e^x \cos y + 2 \cos x.$$

$$M_y = e^x \cos y - 2 \sin x, N_x = e^x \cos y - 2 \sin x. \text{ Exact.}$$

$$F = \int M dx = e^x \sin y + 2y \cos x + h(y)$$

$$F_y = e^x \cos y + 2 \cos x + h'(y) = e^x \cos y + 2 \cos x \Rightarrow h'(y) = 0 \Rightarrow h = 0$$

$$\boxed{e^x \sin y + 2y \cos x = C}$$

Example 3: Solve $(3x^2y + 2xy^2) dx + (x^3 + 2x^2y) dy = 0$.

$$M_y = 3x^2 + 4xy, N_x = 3x^2 + 4xy. \text{ Exact.}$$

$$F = \int (3x^2y + 2xy^2) dx = x^3y + x^2y^2 + h(y)$$

$$F_y = x^3 + 2x^2y + h'(y) = x^3 + 2x^2y \Rightarrow h'(y) = 0$$

$$\boxed{x^3y + x^2y^2 = C}$$